

Hackathon Project: AI-Powered Assistant for Automotive Parts & After-Sales Platform

Updated structured project document prepared from the provided hackathon problem statement.

Purpose
Define the problem, solution approach, RAG capabilities, hybrid API usage and expected impact for an AI assistant supporting an automotive parts and after-sales platform.

1. Problem Statement

Automotive parts dealers and service partners rely on a complex multi-module platform to manage their day-to-day operations — covering purchase orders, order tracking, delivery status, billing, claims, returns, and parts inventory. While the platform is feature-rich, it creates significant friction for users.

Business processes are fragmented across multiple screens and modules. A user checking the status of a claim, for example, must navigate through order tracking, delivery status, and billing screens before getting a complete picture. Information needed to complete a task is scattered across transactional data, SOPs, user manuals, FAQs, and error catalogs — none of which are easily accessible from within the application itself.

This results in high dependency on support teams for routine process guidance, longer resolution times, and reduced productivity — particularly for newly onboarded dealers who are still learning the system.

2. Proposed Solution

We are building an AI-powered conversational assistant embedded within the platform that gives users a single natural language interface across all modules.

Real-time system intelligence	Calls live backend APIs to retrieve order statuses, delivery information, billing documents, claim details, parts availability, and BOM data on demand.
Business knowledge retrieval	Answers process and application questions using a RAG-based knowledge base built from SOPs, user manuals, FAQs, error catalogs, and business policies.
Self-evolving knowledge base	When the assistant retrieves information from a live API response that is not yet captured in the knowledge repository, it automatically enriches the knowledge base with that new information for future queries without manual documentation effort.

A user can simply ask “Where is my delivery for order PO-2026-00891?” or “How do I submit a quarterly return claim?” and receive an accurate, contextual answer — without raising a support ticket or navigating multiple screens.

3. Expected Impact

- Reduces support ticket volume through self-service
- Faster onboarding for dealers and new users
- Single interface replacing multi-screen navigation
- Knowledge base grows continuously from real usage, reducing documentation gaps over time
- Reusable AI assistant framework extensible to other enterprise modules

4. Query Types and RAG System Capabilities

The assistant operates in two modes. For process, policy, and application guidance it retrieves answers from the RAG knowledge base. For live transactional data it calls backend APIs. Many queries combine both modes.

The knowledge base is built from User Manuals, SOPs, Knowledge Base articles, Business Process documents, Policies, FAQs, and an Error Catalog.

4.1 RAG-Based Knowledge Queries

Answered entirely from the document knowledge base — no API call needed.

Process and How-To

Example Query	
How do I place a spare parts order?	
How do I submit a warranty claim?	
How do I submit a quarterly return?	
How do I find a part if I don't know the part number?	

Sample Interaction	Assistant Response
User: How do I submit a damage claim?	Assistant: Go to Claim Declaration, select damage claim, and provide the delivery reference, affected parts, quantities, and a description of the damage. Photos are recommended. Once submitted, track progress in the Claim Tracking screen.

Policy and Eligibility

Example Query	
How long is the warranty on brake components?	
Can I return electrical parts in the quarterly return?	
Can I cancel an order after it has been submitted?	

Sample Interaction	Assistant Response
User: Can I return electrical parts in the quarterly return?	Assistant: No. Electrical components are excluded from scheduled return programs. Eligible parts must be in original undamaged packaging and must not have been installed or used.

Concepts, Troubleshooting, and Access

Example Query
What is a delivery backlog?

Example Query
What does 'partially delivered' mean?
Why was my order blocked at checkout?
I got an error when placing my order — what does it mean?
Why can I not see the billing section?

Sample Interaction	Assistant Response
User: What is a delivery backlog?	Assistant: A backlog is created when a part cannot be included in the initial shipment due to insufficient stock. The system commits to fulfilling the outstanding quantity once stock is available. Monitor backlog items in the Delivery Backlog screen.

4.2 Hybrid Queries — RAG + Live API

Example Query	What the Agent Does
Where is my delivery for order PO-2026-00891?	Calls getOrderList + getDeliveryList, uses Knowledge Base to explain any exception status
My claim was rejected — can I appeal?	Calls getClaimMonitorList for rejection reason, uses Policies for resubmission guidance
Is part PART-00451-A in stock anywhere?	Calls validateMaterial + getStockByPlant, uses Knowledge Base to explain backlog options if out of stock
I got a credit limit error — what can I do?	Uses Error Catalog for the error, uses Knowledge Base and FAQ for next steps

4.3 RAG vs Live API

The RAG system answers how things work. The API answers what is happening right now.

Query	Handled By
What is the status of my order?	Live API
Show me my invoices from last month	Live API
How do I raise a claim?	RAG
What is the warranty period for electrical parts?	RAG

Note: Source Category columns from the provided tables were intentionally removed, as requested.